Question 1: Consider a FICO pipeline with 4 stages:

IF DISP EX* SU

EX takes 2 (pipelined) cycles for LW, and 1 cycle for ADD. There is only one ADD unit and one LW unit. Two instructions per cycle are fetched by IF, and the DISP queue is unlimited in length.

Show the pipeline timing diagram for the following sequence of instructions as shown in class. You can assume that when as instruction completes and clears the busy bit on its destination register, any instruction, waiting in DISP for that register can fire in the same cycle.

A. LW R1, 0(R2)
B. ADD R3, R1, R2
C. ADD R5, R4, R2
D. LW R6, 0(R4)
E. ADD R7, R1, R5
F. ADD R8, R6, R3

	0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
A									
B									
C									
D									
E									
F									
	9	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**
A									
B									
C									
D									
E									
F									

- **Highlight boxes where an instruction cannot make forward progress due to resource limitations.**

Question 2: The sequence in question 1 is now executed on an out-of-order processor that has six LW units and six ADD units. As before, LW is pipelined into 2 cycles, ADD is a single-cycle operation. The fetch rate is now 6 instructions/cycle. **Show the contents of the scheduling queue for traditional Tomasulo (no RAT/Pregs) and the register file** at two snapshots in time:

a) After all 6 instructions have been dispatched, and
b) After instructions A, C and D have completed.

Assume tags begin at the value 100. The sizes of CDBs and Scheduling Queue are unlimited. After an instruction completes and goes into State Update, we will update the RF of its destination register and any instructions waiting in SCHED for that register can fire in the same cycle. **Use the given tables for your final answer**. Note: You do not need to show the register values, just the tag, ready and register number fields of the scheduling queue.

a) After all 6 instructions have been dispatched

FU	Dest Reg	Dest Reg Tag	Src1 Ready	Src1 Tag	Src2 Ready	Src2 Tag

Register #	Ready	Tag
R1		
R2		
R3		
R4		
R5		
R6		
R7		

R8		
R9		
R10		

b) After instructions A, C and D have completed.

FU	Dest Reg	Dest Reg Tag	Src1 Ready	Src1 Tag	Src2 Ready	Src2 Tag

Register #	Ready	Tag
R1		
R2		
R3		
R4		
R5		
R6		
R7		
R8		
R9		
R10		

Question 3: The following instruction stream is executed on a superscalar pipeline that uses the PReg/RAT approach for register renaming. Show the state of the scheduling queue, register file and RAT after all the instructions have been dispatched:

A. LW R1, 0(R2)
B. ADD R3, R1, R2
C. ADD R5, R4, R2
D. LW R6, 0(R4)
E. ADD R7, R1, R5
F. ADD R8, R6, R3

You can assume that Pregs start from 0.